

Azure App Credential Monitoring

Enterprise-Grade Automated Secret & Certificate Expiry Tracking | Vivek Billore

5,000+

Credentials Tracked

90%

Manual Effort Reduced

Zero

Pipeline Failures

~\$0

Infrastructure Cost

■ Problem Statement

In large enterprises, hundreds of Azure App Registrations (Service Principals) power pipelines and applications. When a secret or certificate silently expires: pipelines fail in production, manual spreadsheet tracking is error-prone, and there is no visibility into credential ownership. This system eliminates all three problems with a **fully automated daily monitoring pipeline**.

■ Architecture Flow

Azure AD Tenant

Source of all App Registrations & Service Principals



Microsoft Graph API (Batch)

Fetches all apps, secrets, certs & owners in bulk — 5x faster than sequential calls



Databricks PySpark Notebooks

Processes credentials: expiry calc, replacement detection via Window Functions, owner mapping



Delta Lake (ADLS Gen2)

Stores daily snapshots: Bronze (raw) → Silver (processed) → Gold (alert payload)



Azure Data Factory

Orchestrates daily pipeline: PL_EDWH_AppRegExpiryAlert — scheduled 6AM UTC



Power Automate

Sends personalized multi-tier HTML email alerts to credential owners



■ Alert Logic

Days to Expiry	Alert Type	Frequency	Action Required
15 days	Warning Email	Once	Plan renewal
1–5 days	Urgent Daily Alert	Daily	Renew immediately
0 (expired)	Expired Notification	Monthly digest	Renew / Investigate
Replaced	Suppressed	Never	No action needed

■ ADF Pipeline Structure

Activity	Description
PL_EDWH_AppRegExpiryAlert	Master Pipeline — daily 6AM UTC schedule
ACT_NB_FetchCredentials	Runs notebook 01 — Graph Batch API → Bronze Delta
ACT_NB_ProcessExpiryLogic	Runs notebook 02 — expiry calc + Window Function replacement detection
ACT_NB_GenerateAlertPayload	Runs notebook 03 — JSON payload + Power Automate HTTP trigger

■ Delta Lake Schema (Medallion Architecture)

Layer	Table Name	Contents
Bronze	app_registrations_snapshot	Raw snapshot: app_id, credential_id, end_date, owner_names, owner_emails, snapshot_date
Silver	credential_expiry_status	Adds: days_to_expiry, expiry_status (HEALTHY/WARNING/CRITICAL/EXPIRED), is_replaced, alert_required
Gold	alert_dispatch_log	Owner email, credential count, dispatch status per run

■ Tech Stack

Component	Technology
Credential Fetching	Microsoft Graph API (Batch)
Data Processing	PySpark, Databricks (Unity Catalog)
Storage	Delta Lake (Bronze/Silver/Gold), ADLS Gen2
Orchestration	Azure Data Factory
Alerting	Power Automate, HTML Email
Secret Management	Databricks Secret Scope

■ Key Technical Highlights

- Graph Batch API — fetches owners for 100 apps in a single HTTP request instead of 100 sequential calls
- PySpark Window Functions — lag/lead detect credential replacements, suppressing redundant alerts
- Recursive team traversal — walks manager hierarchy to find all team members
- Medallion Architecture — Bronze (raw snapshot) → Silver (expiry status) → Gold (alert payload)
- Pipe-separated owners — multi-owner apps stored as 'Name1|Name2' for simple, query-friendly format
- Near-zero cost — runs on existing Databricks + ADF, no additional infrastructure required

Demo version — company-specific data removed | github.com/VivekBillore/azure-app-credential-monitor